

Hospital Management System using Oracle APEX

Oracle APEX Assignment

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1. Introduction

This project presents a Hospital Management System developed using Oracle APEX and an Oracle relational database. Its purpose is to provide a structured environment for managing core hospital information, including departments, doctors, patients, medicines, appointments, clinical visits, prescriptions, room assignments, admissions, and operational reports.

Oracle Application Express (APEX) is a low-code development platform that enables secure, data-driven web applications to be created directly on Oracle Database. It was selected because it combines rapid interface development, SQL integration, authentication, interactive reports, charts, validations, and responsive components in one environment.

A hospital management scenario was chosen because it demonstrates realistic database relationships and workflows. The domain requires both independent CRUD pages and interconnected processes, making it appropriate for demonstrating relational design, lookup fields, master-detail behavior, analytical dashboards, and reporting.

The principal objectives were to design a normalized database, implement reliable data-entry screens, connect related clinical and administrative records, provide clear operational indicators, and deliver an application that is readable, maintainable, and useful to hospital staff.

2. Project Objectives

Create CRUD screens: Interactive forms and grids allow authorized users to create, read, update, and delete hospital records.

Create Master-Detail screens: Related records are displayed together, such as patients with appointments, visits, and admissions.

Create forms with lookup fields: Foreign keys are presented through readable lists of values for departments, doctors, categories, rooms, and statuses.

Create complex multi-table screens: Clinical and administrative workspaces combine information from several related tables.

Create a KPI dashboard: The home page summarizes key activity and visualizes appointments, patients, medicines, visits, and admissions.

3. Database Design

Table 1: Database Table Summary

Table	Description
Departments	Hospital departments and service units.
Doctors	Doctor identity, contact, specialty, and department details.
Doctor Specialties	Clinical specialty lookup values.
Patients	Patient demographic and contact information.
Medicines	Medicine details, price, stock, and reorder information.
Medicine Categories	Medicine classification lookup values.
Appointments	Scheduled patient appointments with doctors and statuses.
Appointment Statuses	Standard appointment workflow statuses.
Patient Visits	Clinical visit findings, diagnosis, notes, and follow-up.
Prescriptions	Prescription headers linked to patient visits.
Prescription Items	Medicines, dosage, frequency, and duration per prescription.
Admissions	Inpatient admission, discharge, doctor, room, and status data.
Rooms	Room numbers, types, and availability.

4. Sample Data

Representative sample data was inserted to test CRUD operations, lookup fields, reports, dashboard indicators, and complete clinical workflows. The implemented volumes meet or exceed the assignment requirements.

Table 2: Sample Data Summary

Table	Required	Implemented
Patients	20	20 — Implemented
Doctors	10	10 — Implemented
Departments	5	5 — Implemented
Medicines	20	20 — Implemented
Appointments	30	30 — Implemented
Patient Visits	10	10 — Implemented
Prescriptions	10	10 — Implemented
Rooms	5	5 — Implemented
Admissions	4	4 — Implemented

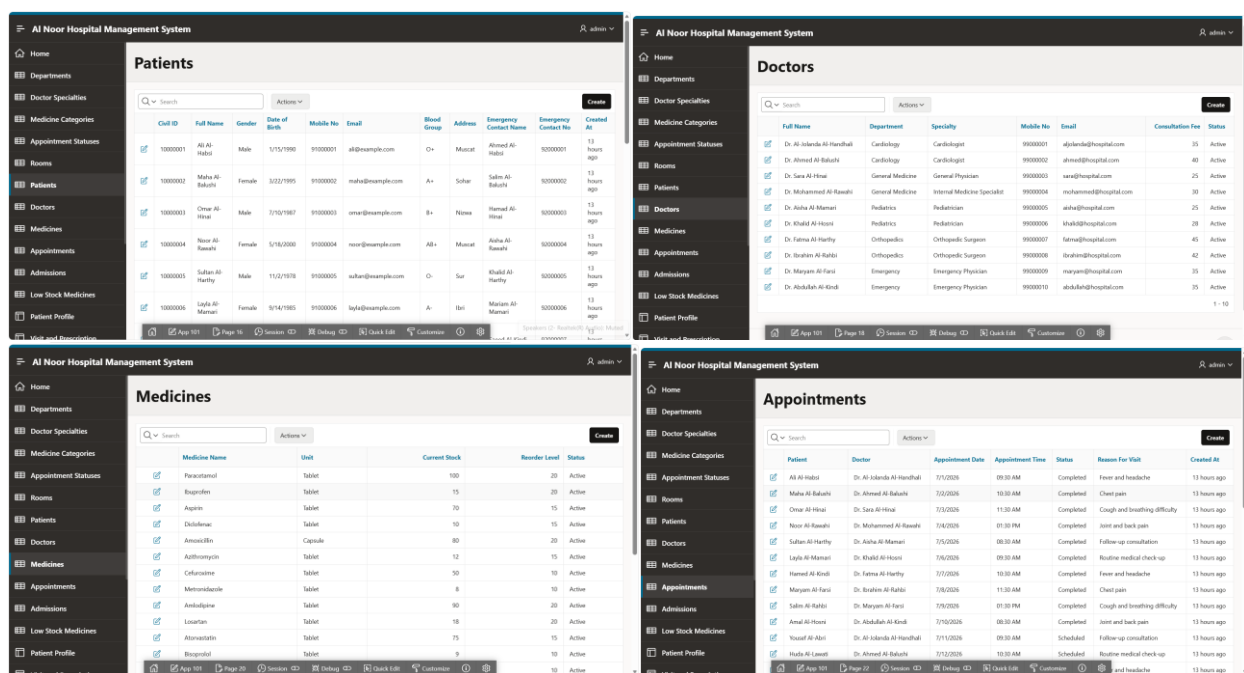


Figure 1: Sample Data Verification Screenshots

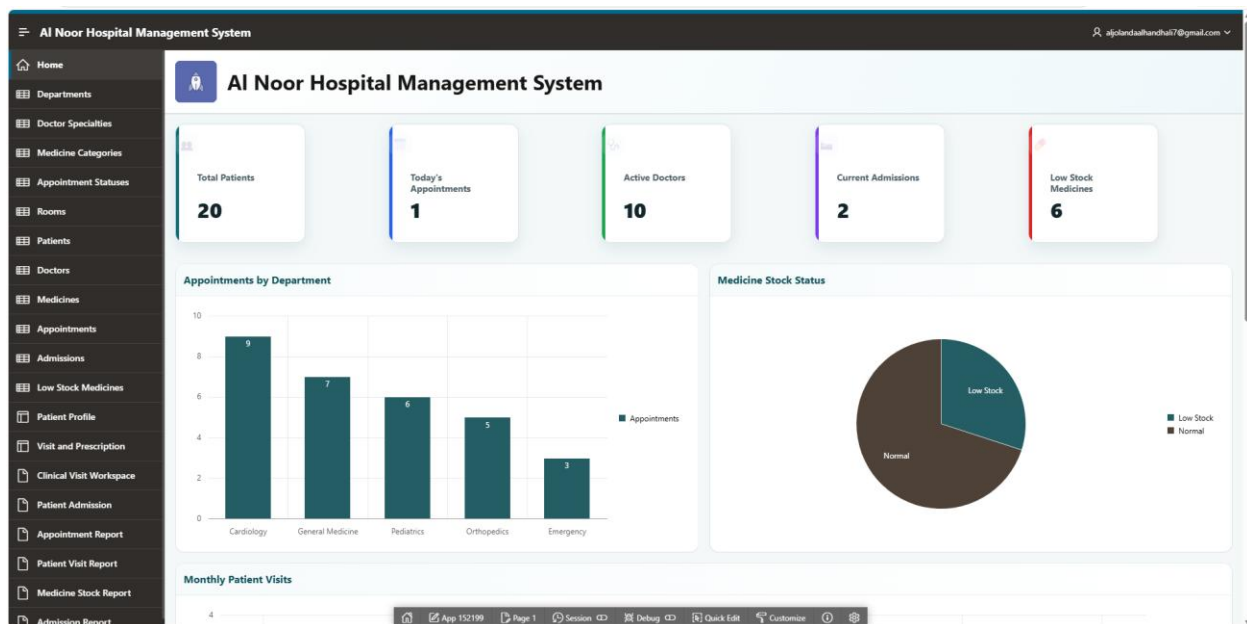
5. Oracle APEX Application

The application is organized around a dashboard, reference-data maintenance pages, clinical transaction pages, multi-table workspaces, and operational reports. Oracle APEX components include Interactive Grids, Interactive Reports, forms, lists of values, dynamic actions, validations, charts, and SQL-based regions.

5.1 Dashboard

The Dashboard provides an immediate summary of hospital activity and supports rapid operational review.

- KPIs: Total Patients, Today's Appointments, Active Doctors, Current Admissions, and Low Stock Medicines.
- Charts: Appointments by Department, Patients by Gender, Medicine Stock Status, Admissions by Room Type, and Monthly Patient Visits.
- SQL queries aggregate live application data, while responsive cards and chart regions present the results clearly.



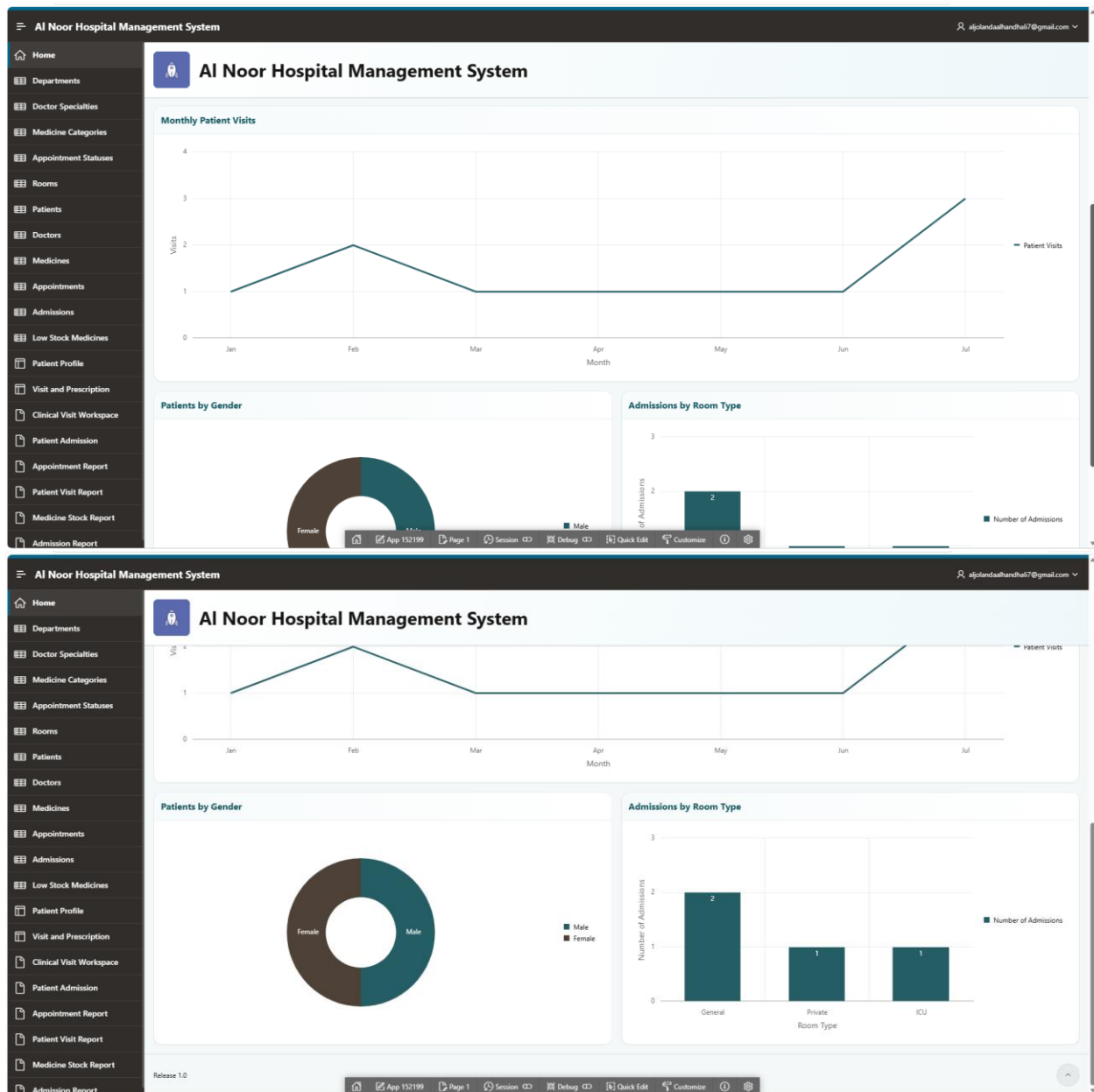


Figure 2: Dashboard Page

5.2 Departments

The Departments page maintains hospital service units and provides complete CRUD functionality.

- Users can add, review, edit, and remove department records.
- Validation prevents incomplete or inconsistent department data.

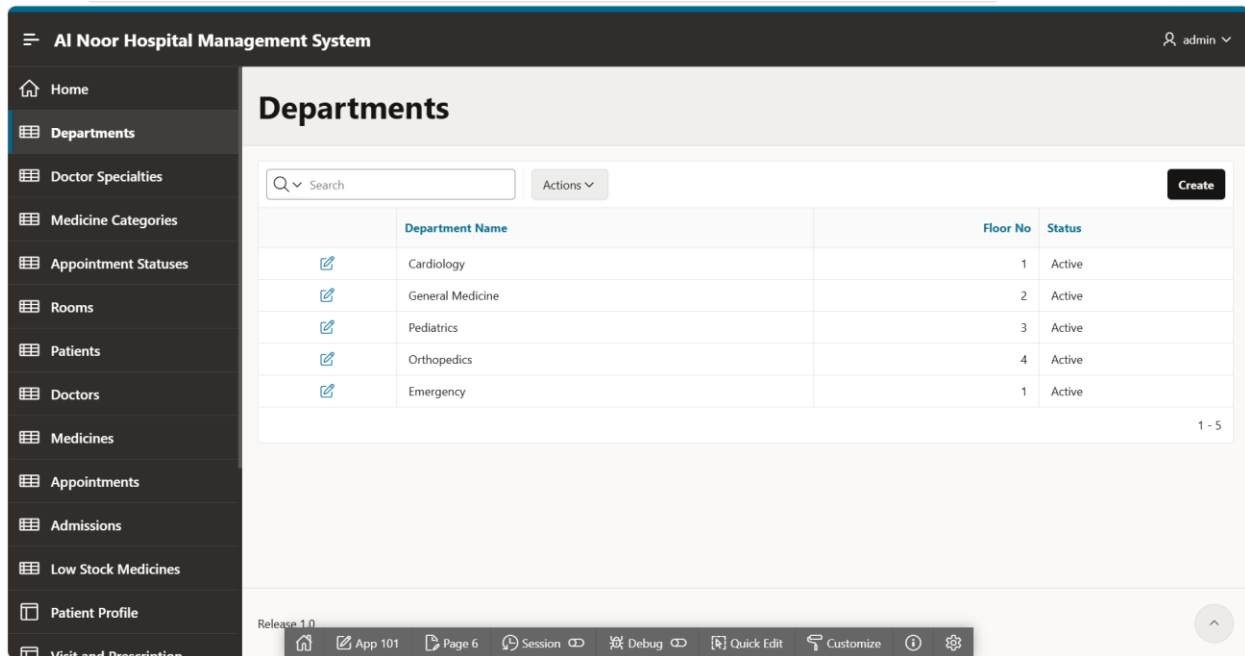


Figure 3: Departments Page

5.3 Doctor Specialties

This reference page maintains the clinical specialties used by doctor records.

- CRUD operations support controlled maintenance of specialty values.
- The lookup structure avoids repeated free-text specialty names.

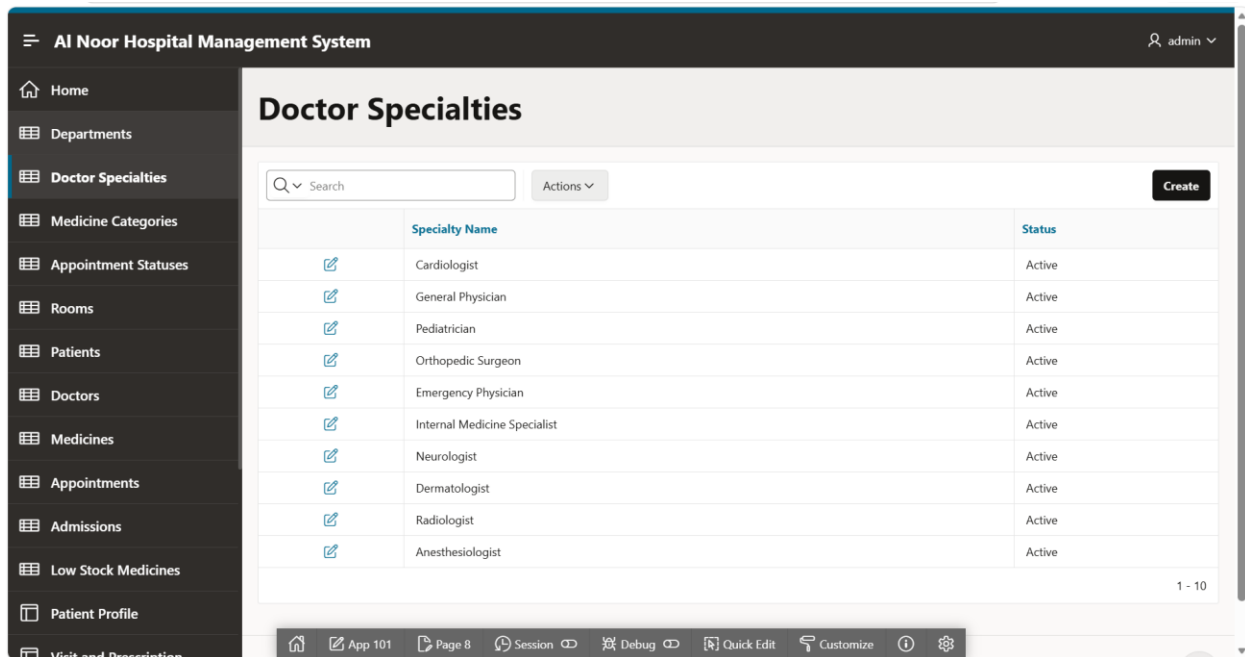


Figure 4: Doctor Specialties Page

5.4 Doctors

The Doctors page manages clinical staff and connects each doctor to organizational and professional lookup data.

- Department and specialty fields use lookup lists rather than raw identifiers.
- Foreign-key relationships support accurate filtering and reporting.

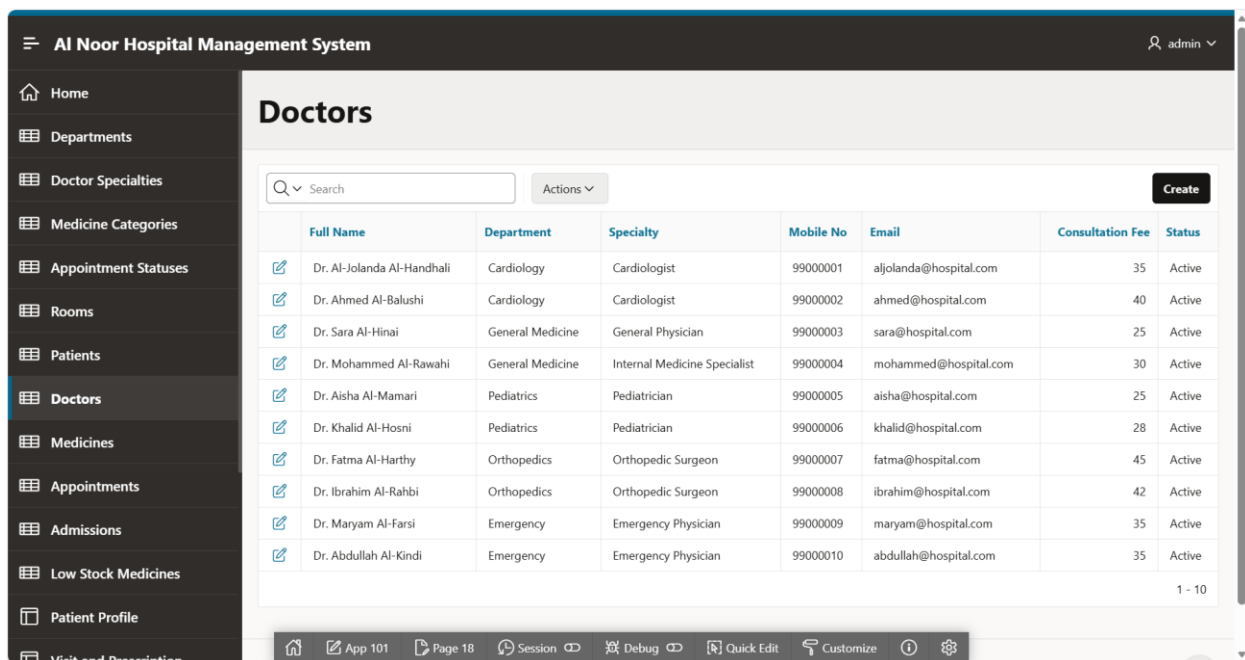


Figure 5: Doctors Page

5.5 Patients

The Patients page registers and maintains patient demographic and contact details.

- CRUD operations support the full patient lifecycle.
- Validations protect required fields, dates, and identifying information.

	Civil ID	Full Name	Gender	Date of Birth	Mobile No	Email	Blood Group	Address	Emergency Contact Name	Emergency Contact No	Created At
	10000001	Ali Al-Habsi	Male	1/15/1990	91000001	ali@example.com	O+	Muscat	Ahmed Al-Habsi	92000001	13 hours ago
	10000002	Maha Al-Balushi	Female	3/22/1995	91000002	maha@example.com	A+	Sohar	Salim Al-Balushi	92000002	13 hours ago
	10000003	Omar Al-Hinai	Male	7/10/1987	91000003	omar@example.com	B+	Nizwa	Hamad Al-Hinai	92000003	13 hours ago
	10000004	Noor Al-Rawahi	Female	5/18/2000	91000004	noor@example.com	AB+	Muscat	Aisha Al-Rawahi	92000004	13 hours ago
	10000005	Sultan Al-Harthy	Male	11/2/1978	91000005	sultan@example.com	O-	Sur	Khalid Al-Harthy	92000005	13 hours ago
	10000006	Layla Al-Mamari	Female	9/14/1985	91000006	layla@example.com	A-	Ibri	Mariam Al-Mamari	92000006	13 hours ago

Figure 6: Patients Page

5.6 Medicines

The Medicines page manages the hospital medicine catalogue and inventory attributes.

- Each medicine is related to a medicine category.
- Stock quantity and reorder-level information support the low-stock dashboard indicator.

AI Noor Hospital Management System admin

Medicines

Search Actions Create

	Medicine Name	Unit	Current Stock	Reorder Level	Status
	Paracetamol	Tablet	100	20	Active
	Ibuprofen	Tablet	15	20	Active
	Aspirin	Tablet	70	15	Active
	Diclofenac	Tablet	10	15	Active
	Amoxicillin	Capsule	80	20	Active
	Azithromycin	Tablet	12	15	Active
	Cefuroxime	Tablet	50	10	Active
	Metronidazole	Tablet	8	10	Active
	Amlodipine	Tablet	90	20	Active
	Losartan	Tablet	18	20	Active
	Atorvastatin	Tablet	75	15	Active
	Bisoprolol	Tablet	9	10	Active

App 101 Page 20 Session Debug Quick Edit Customize

Figure 7: Medicines Page

5.7 Appointments

The Appointments page schedules patient consultations and connects patients with doctors and workflow statuses.

- Master-detail behavior keeps appointments associated with the selected patient.
- Lookup fields display readable patient, doctor, and status values.
- Status information supports operational monitoring and reports.

	Patient	Doctor	Appointment Date	Appointment Time	Status	Reason For Visit	Created At
	Ali Al-Habsi	Dr. Al-Jolanda Al-Handhali	7/1/2026	09:30 AM	Completed	Fever and headache	13 hours ago
	Maha Al-Balushi	Dr. Ahmed Al-Balushi	7/2/2026	10:30 AM	Completed	Chest pain	13 hours ago
	Omar Al-Hinai	Dr. Sara Al-Hinai	7/3/2026	11:30 AM	Completed	Cough and breathing difficulty	13 hours ago
	Noor Al-Rawahi	Dr. Mohammed Al-Rawahi	7/4/2026	01:30 PM	Completed	Joint and back pain	13 hours ago
	Sultan Al-Harthy	Dr. Aisha Al-Mamari	7/5/2026	08:30 AM	Completed	Follow-up consultation	13 hours ago
	Layla Al-Mamari	Dr. Khalid Al-Hosni	7/6/2026	09:30 AM	Completed	Routine medical check-up	13 hours ago
	Hamed Al-Kindi	Dr. Fatma Al-Harthy	7/7/2026	10:30 AM	Completed	Fever and headache	13 hours ago
	Maryam Al-Farsi	Dr. Ibrahim Al-Rahbi	7/8/2026	11:30 AM	Completed	Chest pain	13 hours ago
	Salim Al-Rahbi	Dr. Maryam Al-Farsi	7/9/2026	01:30 PM	Completed	Cough and breathing difficulty	13 hours ago
	Amal Al-Hosni	Dr. Abdullah Al-Kindi	7/10/2026	08:30 AM	Completed	Joint and back pain	13 hours ago
	Yousef Al-Abri	Dr. Al-Jolanda Al-Handhali	7/11/2026	09:30 AM	Scheduled	Follow-up consultation	13 hours ago
	Huda Al-Lawati	Dr. Ahmed Al-Balushi	7/12/2026	10:30 AM	Scheduled	Routine medical check-up	13 hours ago

Figure 8: Appointments Page

5.8 Patient Profile

The Patient Profile page combines multiple related datasets to provide one consolidated view of a patient.

- Patient Profile: demographic and contact information.
- Appointment History: scheduled and completed appointments.
- Visit History: clinical visits, symptoms, diagnosis, notes, and follow-up.
- Admission History: inpatient admission, room, doctor, reason, and status information.
- The master-detail concept passes the selected patient identifier to dependent regions and refreshes them to show related records.

AI Noor Hospital Management System

admin

Home

Departments

Doctor Specialties

Medicine Categories

Appointment Statuses

Rooms

Patients

Doctors

Medicines

Appointments

Admissions

Low Stock Medicines

Patient Profile

Visit and Prescription

Patient Profile

Save

Patient Profile

Search: All Text Columns

Actions

Edit

Add Row

Reset

		Civil Id	Full Name	Gender	Date Of Birt	Mobile No	Email	Blood Group	Address	Emergency C	Emergency C	Created At
☒		10000001	Ali Al-Ha...	Male	1/15/1990	91000001	ali@exam...	O+	Muscat	Ahmed A...	92000001	7/16/2026
☐		10000002	Maha Al-...	Female	3/22/1995	91000002	maha@e...	A+	Sohar	Salim Al-...	92000002	7/16/2026
☐		10000003	Omar Al-...	Male	7/10/1987	91000003	omar@ex...	B+	Nizwa	Hamad A...	92000003	7/16/2026
☐		10000004	Noor Al-...	Female	5/18/2000	91000004	noor@ex...	AB+	Muscat	Aisha Al-...	92000004	7/16/2026
☐		10000005	Sultan Al-...	Male	11/2/1978	91000005	sultan@e...	O-	Sur	Khalid Al-...	92000005	7/16/2026
☐		10000006	Layla Al-...	Female	9/14/1985	91000006	layla@ex...	A-	Ibri	Mariam A...	92000006	7/16/2026
☐		10000007	Hamed A...	Male	6/25/1992	91000007	hamed@...	B-	Muscat	Saeed Al-...	92000007	7/16/2026
☐		10000008	Maryam ...	Female	12/12/19...	91000008	maryam...	O+	Salalah	Ahmed A...	92000008	7/16/2026
☐		10000009	Salim Al-...	Male	4/8/1968	91000009	salim@ex...	A+	Rustaq	Abdullah ...	92000009	7/16/2026

App 101Page 30SessionDebugQuick EditCustomizeTotal 20

Patient Profile

Save

Patient Profile

Appointment History

Search: All Text Columns

Actions

Edit

Add Row

Reset

		Doctor Id	Appointment Date	Appointment Time	Status	Reason For Visit	Created At
☒		21	7/1/2026	09:30 AM	Completed	Fever and headache	7/16/2026
☐		21	7/21/2026	09:30 AM	Cancelled	Cough and breathing ...	7/16/2026

1 rows selectedTotal 2

Visit History

5.9 Visit and Prescription

This page records clinical outcomes and medication orders as one connected process.

- Visit Details store symptoms, diagnosis, notes, and follow-up information.
- Prescription records are linked to visits.
- Prescription Items identify medicines, dosage, frequency, and duration.

Al Noor Hospital Management System admin

Visit and Prescription

Save

▼ Patient Visit Information

Search: All Text Columns Actions Edit Add Row Reset

		Appointment Id..	Patient	Doctor	Visit Date	Symptoms	Diagnosis	Notes	Follow Up Date
☒		41	Ali Al-Habsi	Dr. Al-Jolanda ...	7/10/2026	Fever and cough	Influenza	Patient examin...	7/16/2026
☐		42	Maha Al-Balushi	Dr. Ahmed Al-...	1/15/2026	Chest discomfort	Hypertension	Patient examin...	7/17/2026
☐		43	Omar Al-Hinai	Dr. Sara Al-Hinai	2/12/2026	Shortness of br...	Asthma	Patient examin...	7/18/2026
☐		44	Noor Al-Rawahi	Dr. Mohammed...	3/18/2026	Joint pain	Muscle strain	Patient examin...	7/19/2026
☐		45	Sultan Al-Harthy	Dr. Aisha Al-M...	4/20/2026	Headache and ...	Migraine	Patient examin...	7/20/2026
☐		46	Layla Al-Mamari	Dr. Khalid Al-H...	5/8/2026	Fever and cough	Influenza	Patient examin...	7/21/2026
☐		47	Hamed Al-Kindi	Dr. Fatma Al-H...	6/25/2026	Chest discomfort	Hypertension	Patient examin...	7/22/2026
☐		48	Maryam Al-Farsi	Dr. Ibrahim Al-...	7/10/2026	Shortness of br...	Asthma	Patient examin...	7/23/2026
☐		49	Salim Al-Rahbi	Dr. Maryam Al-...	1/15/2026	Joint pain	Muscle strain	Patient examin...	7/24/2026

App 101 Page 32 Session Debug Quick Edit Customize Total 10

Al Noor Hospital Management System admin

Visit and Prescription

▼ Prescription Information

Search: All Text Columns Actions Edit Add Row Reset

		Patient	Doctor	Prescription Date	Notes
☒		Ali Al-Habsi	Dr. Al-Jolanda Al-Handhali	7/1/2026	Take medicines as instructed and atte...

1 rows selected Total 1

▼ Prescribed Medications

Search: All Text Columns Actions Edit Save Add Row Reset

		Medicine	Dosage	Frequency	Duration (Days)	Instructions
☒		Paracetamol	500 mg	Twice daily	5	Take after meals
☐		Ibuprofen	1 tablet	Once daily	7	Take with water

1 rows selected Total 2

App 101 Page 32 Session Debug Quick Edit Customize

Figure 10: Visit and Prescription Page

5.10 Clinical Visit Workspace

The Clinical Visit Workspace supports the complete consultation workflow from patient context to treatment documentation.

- The user reviews the appointment and patient context.
- Clinical observations are recorded.
- A prescription and its medicine items can be created without leaving the workflow.

The screenshot displays the 'Clinical Visit Workspace' within the 'Al Noor Hospital Management System'. The interface features a dark sidebar on the left with a menu containing items like Patients, Doctors, Medicines, Appointments, Admissions, Low Stock Medicines, Patient Profile, Visit and Prescription, Clinical Visit Workspace (highlighted), Patient Admission, Appointment Report, Patient Visit Report, Medicine Stock Report, and Admission Report. The main content area is titled 'Clinical Visit Workspace' and includes two dropdown menus for 'Patient Name' and 'Appointment Id'. Below these are several expandable sections: Patient Summary, Appointment Summary, Doctor Summary, Visit Details, Prescription, and Prescription Items. At the bottom, a toolbar contains icons for home, app (App 101), page (Page 2), session, debug, quick edit, customize, and help.

Figure 11: Clinical Visit Workspace

5.11 Patient Admission

The Patient Admission page manages inpatient stays and allocation details.

- Admissions link patients with responsible doctors and rooms.
- Room lookup values improve accuracy and availability management.
- Admission and discharge dates, reason, and status describe the inpatient lifecycle.

The screenshot displays the 'Patient Admission' page within the 'AI Noor Hospital Management System'. The interface features a dark sidebar on the left with a menu containing items like 'Patients', 'Doctors', 'Medicines', 'Appointments', 'Admissions', 'Low Stock Medicines', 'Patient Profile', 'Visit and Prescription', 'Clinical Visit Workspace', 'Patient Admission' (highlighted), 'Appointment Report', 'Patient Visit Report', 'Medicine Stock Report', and 'Admission Report'. The main content area is titled 'Patient Admission' and contains a form under the 'Admission Details' section. This form includes dropdown menus for 'Patient', 'Doctor', and 'Available Room', a date picker for 'Admission Date', and a text area for 'Admission Reason'. At the bottom of the form, there is a 'Discharge Date' field. The footer of the application shows a status bar with icons for home, application (App 101), page (Page 5), session, debug, quick edit, customize, and information.

Figure 12: Patient Admission Page

5.12 Reports

The reporting module provides searchable, sortable, and filterable operational views. Users can apply column filters, search terms, sorting, and export options to focus on relevant records.

5.12.1 Appointment Report

Summarizes scheduled appointments, doctors, patients, dates, and statuses.

5.12.2 Patient Visit Report

Presents clinical visits and supports filtering by patient, doctor, date, and diagnosis.

Al Noor Hospital Management System admin

Patient Visit Report

Patient Visit Filters

Doctor: - All Doctors -

Department: - All Departments -

From Visit Date:

To Visit Date:

Apply Filters

Clear Filters

Patient Visit Report

Patient Name	Doctor Name	Visit Date	Symptoms	Diagnosis	Follow Up Date
Ali Al-Habsi	Dr. Al-Jolanda Al-Handhali	7/10/2026	Fever and cough	Influenza	7/16/2026
Maryam Al-Farsi	Dr. Ibrahim Al-Rahbi	7/10/2026	Shortness of breath	Asthma	7/23/2026
Hamed Al-Kindi	Dr. Fatma Al-Harthy	6/25/2026	Chest discomfort	Hypertension	7/22/2026
Layla Al-Mamari	Dr. Khalid Al-Hosni	5/8/2026	Fever and cough	Influenza	7/21/2026
Sultan Al-Harthy	Dr. Aisha Al-Mamari	4/20/2026	Headache and tiredness	Migraine	7/20/2026
Noor Al-Rawahi	Dr. Mohammed Al-Rawahi	3/18/2026	Joint pain	Muscle strain	7/19/2026
Omar Al-Hinai	Dr. Sara Al-Hinai	2/12/2026	Shortness of breath	Asthma	7/18/2026
Amal Al-Hosni	Dr. Abdullah Al-Kindi	2/12/2026	Headache and tiredness	Migraine	7/25/2026
				Hypertension	7/17/2026

Figure 14: Patient Visit Report

5.12.3 Medicine Stock Report

Highlights inventory quantities and medicines at or below reorder level.

The screenshot displays the 'Medicine Stock Report' in the AI Noor Hospital Management System. The interface features a sidebar with navigation options, a top header, and a main content area with filters and a table of medicine stock data.

Medicine Stock Filters

- Category: - All Categories -
- Stock Status: - All Statuses -
- Buttons: Apply Filters, Clear Filters

Medicine Stock Report

Medicine Name	Category	Current Stock	Reorder Level	Stock Status
Amlodipine	Cardiovascular	90	20	Normal
Amoxicillin	Antibiotics	80	20	Normal
Aspirin	Pain Relief	70	15	Normal
Atorvastatin	Cardiovascular	75	15	Normal
Azithromycin	Antibiotics	12	15	Low Stock
Bisoprolol	Cardiovascular	9	10	Low Stock
Budesonide	Respiratory	6	10	Low Stock
Cefuroxime	Antibiotics	50	10	Normal
			15	Normal

Figure 15: Medicine Stock Report

5.12.4 Admission Report

Reports inpatient admissions, rooms, doctors, dates, and status.

Al Noor Hospital Management System admin

Admission Report

Admission Filters

Department
- All Departments -

Doctor
- All Doctors -

Status
- All Statuses -

Apply Filters
Clear Filters

Admission Report

Release 1.0

App 101 Page 34 Session Debug Quick Edit Customize

Al Noor Hospital Management System admin

Admission Report

Admission Filters

Admission Report

Search Actions

Patient Name	Doctor Name	Room Number	Room Type	Admission Date	Discharge Date	Status
Maha Al-Balushi	Dr. Sara Al-Hinai	201	Private	7/12/2026		Admitted
Ali Al-Habsi	Dr. Al-Jolanda Al-Handhali	101	General	7/10/2026		Admitted
Noor Al-Rawahi	Dr. Maryam Al-Farsi	202	Private	6/20/2026	6/21/2026	Discharged
Omar Al-Hinai	Dr. Fatma Al-Harthy	102	General	6/15/2026	6/18/2026	Discharged

1 - 4

Release 1.0

App 101 Page 34 Session Debug Quick Edit Customize

Figure 16: Admission Report

6. Application Flow

A typical user journey follows a controlled sequence from authentication and reference-data setup to clinical transactions and reporting. The flow ensures that lookup records exist before dependent operational data is created.

1. User logs in securely.
2. Dashboard opens and displays hospital KPIs and charts.
3. Departments and doctor specialties are maintained.
4. Doctors are added and linked to departments and specialties.
5. Patients are registered and validated.
6. Medicines and medicine categories are maintained.
7. Appointments are scheduled using patient, doctor, and status lookups.
8. Clinical visits are recorded against the patient and appointment context.
9. Prescriptions and prescription items are created.
10. Patients are admitted and assigned to rooms when required.
11. Interactive reports are filtered, reviewed, and exported.

7. Conclusion

The Hospital Management System demonstrates how Oracle APEX can be used to build a complete, database-centered web application within a short development cycle. The project strengthened practical experience in relational database design, SQL, Oracle APEX page development, validations, dynamic actions, and user-oriented interface design.

The final application integrates CRUD operations, lookup fields, master-detail relationships, multi-table clinical workflows, dashboard KPIs, charts, and interactive reporting. Together, these features provide a coherent solution for managing hospital reference data and operational activities. The work also illustrates the importance of referential integrity, reliable session-state handling, clear navigation, and maintainable page structures in professional software engineering projects.